

COMPUTE ALLOCATION REPORT

University HPC Centers

The fastest win — your own campus cluster, with a deep-dive on USC Hyperion

J.C. Vaught · April 2026

Why This Matters First

For most graduate students, the campus HPC center is the fastest, lowest-friction compute pathway on the planet. Hardware is already paid for out of indirect-cost recovery, capital reserves, or institutional research funds. There is no competitive review, no merit panel, no narrative justification beyond an advisor's signature. Onboarding is typically same-day or next-day. Wall-clock time from “I should learn Slurm” to “my first job is queued on an A100” is routinely under 48 hours.

This report argues that any grad student building a compute strategy should investigate their home institution's HPC offering *before* writing a single federal proposal. NAIRR, ACCESS, INCITE, and ALCC are powerful multipliers, but they are slower, more competitive, and more administratively heavy. A campus cluster covers exploration, debugging, baseline runs, ablations, and the long tail of small experiments that dominate the actual research calendar. Federal allocations are then reserved for the few large-scale runs that truly need them.

The same logic applies to industry credit programs. Google TRC, AWS Cloud Credits for Research, and Azure for Researchers are excellent, but every hour spent debugging a Docker container on a rented spot instance is an hour you could have spent training on a node your tuition already paid for.

The General University-HPC Landscape

Campus HPC centers in the United States cluster around three operational models. Understanding which model your institution uses determines how you ask for time, how much you get, and how predictable your queue waits are.

The *condo model* is dominant at large land-grant and R1 systems. Faculty PIs purchase nodes out of grant funds, and those nodes are integrated into the central cluster. The PI gets priority access to the equivalent of what they bought, and the broader user community gets cycle-scavenging access to idle condo nodes through a low-priority preemptable queue. Examples include Clemson Palmetto2, Purdue's community clusters, and Iowa State's Nova.

The *open or institutional model* funds a shared pool centrally. Any researcher with an advisor sponsor gets fair-share access. There is no buy-in, but priority is uniform and queue depth at peak times can be painful. Many smaller R1s and most R2s use this model.

The *hybrid model* is the most common in practice. A central institutional pool covers everyone, condo nodes layer on top for funded labs, and a small “AI partition” or “GPU partition” is funded out of a recent NSF MRI, NIH S10, or internal strategic-investment award. USC's current posture fits this hybrid pattern, with Hyperion as the institutional backbone and Theia as the strategic AI add-on.

Typical GPU Tiers in 2026

Mid-size R1 campus clusters in the 2024–2026 refresh cycle now commonly carry some mix of NVIDIA A100 (40 or 80 GB), H100 (80 GB), L40S (48 GB), and RTX 6000 Ada (48 GB). H200 and B200 nodes are starting to appear at the largest campuses but remain rare. Older V100 and P100 nodes from the 2018–2020 era are still in service at most sites and are perfectly adequate for development, debugging, and small fine-tunes.

Software, Storage, and Liaisons

Slurm is the de facto scheduler at almost every U.S. campus HPC. PBS Pro survives at a handful of sites. Storage is universally tiered into a small backed-up *home* directory (typically 25–100 GB), a large unbacked-up *scratch* space with a purge policy (typically 1–10 TB per user, purged after 30–90 days of inactivity), and a *project* or *group* space that is either purchased or allocated by the center. A *Globus* endpoint is standard for moving data in and out.

The NSF-funded *Campus Champions* program embeds a local liaison at each member institution who can fast-track the user to ACCESS allocations and broker introductions to other regional resources. Checking whether your campus is a Campus Champion site is a 30-second exercise that often saves weeks of proposal-writing later.

USC Research Computing — Hyperion and Theia

USC Research Computing sits inside the Division of Information Technology and operates three production clusters as of the 2025–2026 academic year. The flagship general-purpose system is *Hyperion*, the new AI-focused system is *Theia*, and the teaching-only cluster is *Bolden*. All three are managed under Bright Cluster Manager and scheduled with Slurm.

Hyperion (Phase III, in production since 2023)

Hyperion Phase III provisions 356 nodes totaling 16,616 CPU cores and a peak of approximately 1.5 PFLOPS. The mix is roughly 295 standard compute nodes, 44–45 GPU nodes, and 8 big-memory nodes. Compute and GPU nodes carry 128–256 GB of system RAM, big-memory nodes carry 2 TB. The interconnect is EDR InfiniBand at 100 Gb/s, a step below the NDR-400 fabric used on Theia.

The GPU complement on Hyperion includes DGX-style nodes with 8x A100 GPUs and older dual-V100 nodes. Storage consists of 450 TB of GPFS home space and 1.4 PB of GPFS scratch. Two dedicated data-transfer nodes serve as Globus endpoints. Specific GPU partition names, per-user quota numbers, and condo priority weights are not enumerated on the public RCI pages and should be confirmed by emailing rci@mailbox.sc.edu at account-creation time.

Theia (NSF MRI-funded, AI-focused)

Theia is the newer, AI-targeted system, funded by an NSF Major Research Instrumentation award to expand AI access across South Carolina institutions. The published configuration is 28 CPU nodes at 112 cores each, 10 quad-GPU nodes (nine quad-A100 nodes and one quad-H100 node), and 1.5 PB of GPFS scratch over a 400 Gb/s NDR InfiniBand fabric. Aggregate is 3,808 CPU cores and approximately 675 combined CPU+GPU TFLOPS at peak. Theia is the recommended target for any modern AI/ML workload at USC.

Bolden (teaching)

Bolden is a 20-node teaching cluster with 460 cores and 300 TB of Lustre scratch. It is reserved for course use and is not a research target.

Support and Training

The RCI team runs periodic Slurm and HPC-onboarding sessions through DoIT and the AI Institute, and offers consulting hours by appointment. The general contact for accounts, allocations, and consulting is rci@mailbox.sc.edu. Faculty in CSE, the AI Institute, and the College of Engineering and Computing have standing relationships with the RCI staff and can usually broker an introduction.

Hardware Specs at a Mid-Size R1

The table below summarizes the GPU types most likely to appear on a mid-size R1 campus cluster in 2026. The FP16/BF16 column is dense tensor throughput without sparsity, directly comparable to the 989 TFLOPS H200 baseline used elsewhere in this report series. The right-hand column is a typical node count of that GPU type at a mid-size R1, useful for estimating queue depth.

GPU model	VRAM (GB)	FP16 TFLOPS	BW (TB/s)	Typical nodes
V100 SXM2	32	125	0.90	4–12
RTX 6000 Ada	48	364	0.96	2–8
L40S	48	362	0.86	2–8
A100 40GB	40	312	1.55	4–16
A100 80GB	80	312	2.04	2–12
H100 SXM5	80	989	3.35	1–8
H200 SXM5	141	989	4.80	0–2

USC's current footprint slots into this picture at the A100-heavy end. Theia's quad-A100 nodes plus its single quad-H100 node are well-aligned with what a typical mid-size R1 carries, with Hyperion's DGX A100 chassis and V100 nodes filling out the older tier.

Allocation Model and Throughput Estimate

Campus clusters almost never price allocations in service-unit hours the way ACCESS or INCITE do. Instead, they allocate by Slurm fair-share weights and partition membership. A new grad student joining an unfunded lab gets baseline fair-share, can submit to any open partition, and is preempted on condo partitions when the owning PI submits. A student in a funded lab inherits the PI's fair-share weight and any condo priority the PI has purchased.

For the purpose of this report series, the question is realistic *steady-state throughput* over a 90-day semester. A new USC student on Theia or Hyperion can plausibly hold 1–2 A100 GPUs continuously without preemption, with bursts to 4–8 A100s for shorter runs. The table below converts steady-state occupancy into H200-hour equivalents using the 312 TFLOPS A100 vs 989 TFLOPS H200 ratio (approximately 0.316 H200-hour per A100-hour).

Steady GPUs	GPU type	A100-hours / 90 days	H200-hour equiv
1	A100 40/80GB	2,160	681

Steady GPUs	GPU type	A100-hours / 90 days	H200-hour equiv
2	A100 40/80GB	4,320	1,363
4	A100 40/80GB	8,640	2,725
8	A100 40/80GB	17,280	5,450
4	H100 80GB	8,640	8,640
8	H100 80GB	17,280	17,280

The two A100-row example matches the canonical estimate. Steady 2x A100 over 90 days yields $90 \times 24 \times 2 \times 312 / 989$ approximately 1,362 H200-hours, an allocation envelope that is comparable to a small NAIRR Start-Up grant delivered with zero proposal overhead.

Approval Timeline

Account creation at USC RCI is form-driven. The user submits the account request through the DoIT Research Computing portal, the listed advisor or PI approves by email, the user completes a short training module on cluster etiquette and Slurm basics, and SSH keys are provisioned. End-to-end turnaround is typically 1–3 business days for a routine request and rarely exceeds a week. There is no committee, no scoring rubric, and no narrative proposal.

Eligibility

Any USC researcher with an active appointment and an advisor sponsor is eligible. This includes graduate students, postdocs, undergraduate research assistants, and visiting scholars. The default partitions on Hyperion and Theia carry no per-user cost. Priority partitions tied to condo investments require the owning PI to add the user to their group. Course-only access on Bolden is granted to enrolled students through the instructor.

Things to Know

Campus storage quotas tend to bite well before compute does. Home directories in the 25–100 GB range fill quickly with conda environments, model checkpoints, and pip caches. Plan to keep working data on scratch and to set up a project-space request early if the workflow exceeds 1–2 TB.

GPU-partition queue waits at peak times such as paper-deadline season can run from hours to several days. The unwritten rule at most campus clusters is to develop and debug interactively on a single GPU, then submit batch arrays during off-peak windows.

USC’s status as an ACCESS Campus Champion site is worth verifying with RCI directly. If USC is a member, the local liaison can fast-track an ACCESS Explore allocation, which is the natural next step after exhausting campus fair-share. Combining campus HPC with NAIRR Start-Up and AFMR credits is the recommended stack for a grad student in 2026, with each layer covering a different scale of run.

Campus IT often subsidizes a quota of consulting hours with RCI staff. Use them early. A 30-minute conversation about how Slurm is actually configured locally saves weeks of trial-and-error. Some campuses also run a graduate seed-grant program that bundles a small compute allocation with mentoring; checking the AI Institute and the Office of the Vice President for Research for current calls is worth a single email.

Strategic Fit

The campus HPC is the first compute pathway any USC grad student should exercise. Zero competitive overhead, immediate access, and a moderate but real H200-hour-equivalent throughput over a semester make it the workhorse layer in any sensible compute strategy. Use Hyperion and Theia for everything that fits, and reserve external programs covered elsewhere in this report series for the runs that genuinely outgrow the campus envelope.

Summary Table

The scenarios below assume a single grad-student user on USC Theia or Hyperion, sustaining the listed concurrency over a 90-day semester. Effort is the wall-clock cost of obtaining and maintaining the allocation, not the science work itself.

Scenario	GPU type	H200-hr equiv / sem	Approval time	Effort
1 GPU steady	A100 40/80GB	681	1–3 days	Trivial
2 GPU steady	A100 40/80GB	1,363	1–3 days	Trivial
4 GPU steady	A100 40/80GB	2,725	1–3 days	Low
8 GPU steady	A100 40/80GB	5,450	1–3 days + condo	Low–Mod.
4 GPU steady	L40S 48GB	3,160	1–3 days	Low
4 GPU steady	H100 80GB	8,640	1–3 days + queue	Low
8 GPU steady	H100 80GB	17,280	1–3 days + condo	Moderate

H200-hour equivalence is computed at dense FP16/BF16 tensor throughput (989 TFLOPS, 141 GB HBM3e, 4.8 TB/s) per H200 SXM GPU. Conversions below use the per-hour FP16 TFLOP ratio; memory-bound workloads scale differently and the bandwidth column is provided so readers can re-derive their own equivalence.

Sources

Verified April 2026 via WebSearch and WebFetch against USC DoIT Research Computing pages and adjacent USC sources.

- USC DoIT Research Computing, Resources page. https://sc.edu/about/offices_and_divisions/division_of_information_technology/rc/resources/
- USC DoIT Research Computing, Hyperion Phase III in production (2023). https://sc.edu/about/offices_and_divisions/division_of_information_technology/rc/news/2023/phasethree_hyperion.php

- USC DoIT Research Computing, news index. https://sc.edu/about/offices_and_divisions/division_of_information_technology/rc/news/
- USC DoIT Research Computing, landing page. https://sc.edu/about/offices_and_divisions/division_of_information_technology/rc/index.php
- USC College of Engineering and Computing, AI Institute Facilities. <https://research.cec.sc.edu/aifacilities>
- USC CSE, NSF MRI award announcement (Theia). <https://www.cse.sc.edu/research/news/usc-awarded-nsf-mri-grant-acquire-hpc-cluster-ai-science-research-and-education-south>
- ACCESS Campus Champions program directory. <https://campuschampions.cyberinfrastructure.org/>
- Primary contact for accounts and allocations: rci@mailbox.sc.edu